

A precision-dependent lower bound for approximately fair dice

Research note; independently reviewed

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Throughout, logarithms are natural, and $\log(1/0) = +\infty$.

Theorem 1. *There is an absolute $c > 0$ with the following property. Let $m \geq 2$, $0 \leq \varepsilon \leq 1/2$, and let $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ be nondecreasing on rows with positive probability weights w_q . Suppose every failure pattern $S \subseteq [m]$ satisfies*

$$\frac{1 - \varepsilon}{(m+1)\binom{m}{|S|}} \leq \mathbb{E}_w \prod_{j \in S} (1 - p_j) \prod_{i \notin S} p_i \leq \frac{1 + \varepsilon}{(m+1)\binom{m}{|S|}}.$$

Then both endpoint row weights are at most

$$\frac{1}{cm \min\{\log m, \log(1/\varepsilon)\}}.$$

In particular, for equal weights,

$$K \geq cm \min\{\log m, \log(1/\varepsilon)\}.$$

Corollary 1. *Suppose every strict order of $n \geq 3$ independent equiprobable finite dice with no cross-die ties has probability in $[(1 - \varepsilon)/n!, (1 + \varepsilon)/n!]$, where $0 \leq \varepsilon \leq 1/2$. Every die then has at least*

$$cn \min\{\log n, \log(1/\varepsilon)\}$$

faces, after changing the absolute constant.

Proof. For any distinguished die and any subset of the other dice, the event that precisely that subset lies above it is a disjoint union of $|S|!(n - 1 - |S|)!$ strict orders. Summing their relative bounds gives the hypothesis of Theorem 1 with $m = n - 1$. $\square$

Proof of Theorem 1. Put $A = \prod_j p_j$ and $\eta(q) = (m+1)w_q A(q)$. This is a positive measure whose mass lies in $[1-\varepsilon, 1+\varepsilon]$; it is not normalized to a probability measure. On $A > 0$ put $r_j = (1-p_j)/p_j$ and $\lambda = \sum_j r_j$, assigning infinite odds to zero entries when defining bad rows. For $B_q(z) = \prod_j (p_j + z(1-p_j))$, the coefficient of z^k in $(m+1)\mathbb{E}_w B_q(z)$ lies in $[1-\varepsilon, 1+\varepsilon]$. For a fixed index j , the coefficient of z^k in

$$(m+1)\mathbb{E}_w \left[(1-p_j) \prod_{i \neq j} (p_i + z(1-p_i)) \right]$$

lies between $(1-\varepsilon)(k+1)/m$ and $(1+\varepsilon)(k+1)/m$.

Take $\tau = m^{-1/2}$, let $D = \{\max_j r_j \geq \tau\}$, and put $G = D^c$. The bad set is an initial segment. If it is nonempty, choose one column witnessing the inequality at its last row. On every bad row, its success probability is at most τ^{-1} times its failure probability. Coefficientwise comparison of the undivided polynomials, which remains valid at zero entries, gives

$$1-\varepsilon-(1+\varepsilon)\frac{(k+1)/\tau+k}{m} \leq a_k := \sum_{q \in G} \eta(q) e_k(r(q)) \leq 1+\varepsilon \quad (0 \leq k < m).$$

Writing $M_k = \sum_{q \in G} \eta(q) \lambda(q)^k / k!$, the collision expansion of ordered index tuples gives

$$0 \leq M_k - a_k \leq \frac{(k-1)\tau}{2} M_{k-1} \quad (k \geq 2), \quad M_1 = a_1.$$

The same induction as in the exact endpoint lemma yields $M_k \leq (1+\varepsilon)(1+k\tau)$ for $k\tau \leq 1$. Combining both estimates,

$$|M_k - 1| \leq \varepsilon + \frac{3(k+1)}{\sqrt{m}} \quad (0 \leq k \leq \sqrt{m}, k < m). \quad (1)$$

No probability renormalization was used in this argument.

We use the positive-measure Laguerre lemma proved in `../size_bounds/logarithmic_per_die_lower_bound.tex`: if normalized moments through degree $2d-1$ have error at most δ and $\delta d 9^d < 1/6$, then some positive mass lies below $2/d$, and any atom at $x \leq 2/d$ has mass at most $32/d + 1024\delta 9^d/d^2$. The lemma explicitly permits a measure whose total mass is not one.

Set

$$H = \min\{\log m, \log(1/\varepsilon)\}, \quad d = \left\lfloor \frac{H}{16 \log 3} \right\rfloor.$$

For sufficiently large H , $d \geq 2$, $2d - 1 \leq \sqrt{m}$, and (1) gives $\delta \leq \varepsilon + 6d/\sqrt{m}$. Since $m \geq e^H$, $\varepsilon \leq e^{-H}$, and $9^d \leq e^{H/8}$,

$$\delta d 9^d \leq d e^{-7H/8} + 6d^2 e^{-3H/8} \longrightarrow 0.$$

The corresponding atom error is $o(1/d)$ as well. Hence the last row is good, has $\lambda(K) < 2/d$, and its η -mass is at most $33/d$, once H exceeds an absolute constant. As $A(K) \geq e^{-\lambda(K)} \geq e^{-2/d}$, this gives $w_K \leq C/(mH)$.

For completeness, the remaining bounded range of H follows from a uniform linear bound. If $m \geq 36$, then

$$\eta(G) \geq 1 - \varepsilon - \frac{1 + \varepsilon}{\sqrt{m}} \geq \frac{1}{4}.$$

At $z = 1/2$, the coefficient bounds give $(m + 1)\mathbb{E}_w B_q(1/2) \leq 3$. On G , $\log(1 + r_j/2) \geq r_j/4$, so

$$\int_G e^{\lambda/4} d\eta \leq 3.$$

Since $e^4/4 > 3$, some good row has $\lambda \leq 16$. The last row therefore has $A(K) \geq e^{-16}$ and

$$(m + 1)w_K e^{-16} \leq \eta(K) \leq 1 + \varepsilon \leq 3/2.$$

This proves $w_K \leq C/m$ with an absolute constant. When $m < 36$, the same estimate follows by increasing that constant, because $w_K \leq 1$. This handles bounded H and completes the last-row bound. Finally reverse the row order and replace every p_j by $1 - p_j$. The failure-pattern hypothesis is unchanged, proving the first-row bound. $\square$

The theorem is a general lower bound for pointwise relative approximation. For fixed nonzero error it is linear in n ; it does not match the known random-palindrome upper bound or establish a superlinear lower bound at fixed accuracy. Its additional factor grows as the requested error shrinks, up to the exact-fairness factor $\log n$.